Write Equations

Lesson 7-1

Name: ______________ Date: ______________ Class: ______________

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

$x + 2 = 7$

balanced with =

$x + 5 = 12$ means 'some number plus 5 equals 12'

Equation

A math sentence with an equal sign showing both sides are the same.

x

stands for a number

In $n + 3 = 10$, the letter n stands for the unknown number ($n = 7$)

Variable

A letter that stands for an unknown number.

$x + 2 = 7$

balanced with =

$7 + 3 = 10$ — both sides equal 10

Equal sign

The = sign, showing both sides are the same.

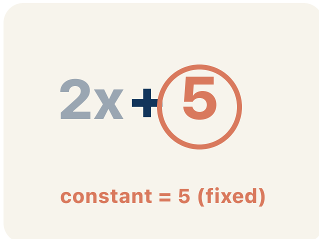
$3x + 5$

no equal sign

$2x + 5$ is an expression; it becomes an equation when you write $2x + 5 = 15$

Expression

A math phrase with numbers and letters, but no equal sign.



In x + 7 = 12, the numbers 7 and 12 are constants; only x can change.

Constant

A fixed number in an expression or equation that does not change.

In n + 5 = 12, n is the unknown; you solve to find n = 7.

Unknown

The value you are trying to find, usually shown with a variable.

Key Ideas & Notes

🎯 **WHAT WE'RE LEARNING TODAY**

I can write an equation to represent a real-world situation.

1️⃣ Notice

Detective Ruiz received a tip: 'A number of stolen gems plus 8 more equals 20 total gems.' She needs to write an equation to represent this clue before she can solve the case.

2️⃣ Key idea

I can write an equation to represent a real-world situation.

3️⃣ Words I'll use

Equation **Variable** **Equal sign** **Expression** **Constant** **Unknown**

🤔 Think About It

- What quantity is unknown in this situation?
- What does 'plus 8 more' tell you about the operation?
- What does 'equals 20 total' tell you?

See the notes in action: watch one worked all the way through, then try the next with the same steps.

👀 I do — watch

Follow each step as your teacher solves it.

Problem: Which equation represents: 'A number plus 12 equals 30'?

A. $n + 12 = 30$

B. $n - 12 = 30$

C. $12n = 30$

D. $n / 12 = 30$

Step 1 'Plus' means addition, so the equation is $n + 12 = 30$.

✅ **Answer:** A. $n + 12 = 30$

🤝 **We do — together**

Solve this one with your class using the same steps.

Problem: Which equation represents: 'Seven less than a number is 18'?

A. $n - 7 = 18$

B. $7 - n = 18$

C. $n + 7 = 18$

D. $n / 7 = 18$

Step 1 ______

Step 2 ______

Answer: ______

✏️ **You do — your turn**

Now try one on your own. Show every step.

Problem: Which equation represents: 'Three times a number equals 21'?

A. $3n = 21$

B. $n + 3 = 21$

C. $n - 3 = 21$

D. $n / 3 = 21$

Show your work:

______

______

______

Turn & Talk — Discussion Points

Level 1 support | Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

Detective Ruiz's clue says 'a number of stolen gems plus 8 more equals 20 total.' How would you turn this clue into an equation, and what does each part stand for?

Level 1 support

Start here: I let a variable stand for the unknown number of gems.

💡 **Need a hint?**

Sentence starters (optional)

The unknown is ____, so I write the variable ____.

La incógnita es ____, así que escribo la variable ____.

My equation is ____ because 'plus 8' means ____ and 'equals 20' means ____.

Mi ecuación es ____ porque 'más 8' significa ____ y 'es igual a 20' significa ____.

Word bank: equation | variable | equal sign | expression

Listen for: *Students choose a variable (such as n) for the unknown gems, build n + 8 = 20, and connect 'plus 8 more' to + 8 and 'equals 20 total' to = 20.*

Level 2

Why does the equation n + 8 = 20 represent the clue better than the expression n + 8 alone? Justify using the role of the equal sign.

- The equation is better than the expression because ____.
- The equal sign tells me ____ that an expression cannot show.

EXPLORE

On the balance scale you put n + 8 on one side and 20 on the other. How did you decide what belongs on each side of the equal sign?

Level 1 support

Start here: Both sides of the equation must represent the same total.

💡 **Need a hint?**

Sentence starters (optional)

I put ____ on the left side because ____.

Puse ____ en el lado izquierdo porque ____.

The equal sign means both sides are ____.

El signo igual significa que ambos lados son ____.

Word bank: equation | variable | equal sign | expression

Listen for: *Students place the expression n + 8 on one side and the total 20 on the other, explaining the equal sign means both sides have the same value.*

Level 2

Compare 'twice a number equals 20' with 'a number plus 8 equals 20.' How would the equations differ, and why?

- 'Twice a number' would be ____ instead of ____ because ____.
- The operation changes the equation because ____.

CONNECT

When would writing an equation help you solve a real problem instead of just guessing?

Level 1 support

Start here: An equation organizes what I know and what I want to find.

💡 **Need a hint?**

Sentence starters (optional)

Writing an equation helps me because ____.

Escribir una ecuación me ayuda porque ____.

I would write an equation to find ____.

Escribiría una ecuación para hallar ____.

Word bank: equation | variable | expression | equal sign

Listen for: *Students explain that an equation captures the relationship clearly, lets you use a variable for the unknown, and can be solved instead of guessing.*

Level 2

Critique: 'Any sentence with numbers is an equation.' Use the words expression and equal sign to explain why this is wrong.

- This is wrong because an expression ____.
- A sentence is only an equation when it has ____.

Write About the Math

The Writing Revolution

I can explain my equation using the words equation, variable, equal sign, and expression.

1. Kernel Sentence

subject + verb

Model: Equation is a math sentence with an equal sign showing both sides are the same.
Ecuación es una oración matemática con un signo igual que muestra que ambos lados son iguales.

Write a kernel sentence about equation. Use a subject and a verb.
Escribe una oración base sobre ecuación. Usa un sujeto y un verbo.

___

___

2. Sentence Expansion

because · but · so

Kernel: Equation matters in math
Ecuación importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because
porque

Equation matters in math because ____.
Ecuación importa en matemáticas porque ____.

___

but
pero

Equation matters in math, but ____.
Ecuación importa en matemáticas, pero ____.

___

so
entonces

Equation matters in math, so ____.
Ecuación importa en matemáticas, entonces ____.

___

3. Sentence Types
4 ways to write a math idea

Statement
Afirmación

Tell one true fact about equation.
Di un hecho verdadero sobre equation.

Equation ____.

Question
Pregunta

Ask a question about equation.
Haz una pregunta sobre equation.

How does ____ ?
¿Cómo ____ ?

Exclamation
Exclamación

Show excitement about equation.
Muestra entusiasmo sobre equation.

Wow, ____ !
¡Guau, ____ !

Command
Mandato

Tell a partner what to do with equation.
Dile a un compañero qué hacer con equation.

First, ____ .
Primero, ____ .

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4. Explain Your Reasoning
use a sentence starter

The unknown is ____, so I write the variable ____.
La incógnita es ____, así que escribo la variable ____.

My equation is ____ because 'plus 8' means ____ and 'equals 20' means ____.
Mi ecuación es ____ porque 'más 8' significa ____ y 'es igual a 20' significa ____.

I put ____ on the left side because ____.
Puse ____ en el lado izquierdo porque ____.

Try It

Solve on your own. Check the answer key when you are done.

1. Which equation represents '3 times a number equals 21'?

A. $3n = 21$

B. $n + 3 = 21$

C. $n/3 = 21$

D. $n - 3 = 21$

Show your work:

2. Tickets cost $8 each. Which equation finds the number of tickets t bought for $48?

A. $8t = 48$

B. $t + 8 = 48$

C. $t/8 = 48$

D. $48t = 8$

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

Write a real-world situation that can be modeled by the equation n / 5 = 9. Explain what n represents and verify your equation makes sense.

Sentence starter: Situation: ____. In this problem, n represents ____. Check: n = ____ because ____ ÷ 5 = 9.

Show your work:

Reflect — Exit Ticket

Which equation represents: 'A number divided by 6 equals 9'?

A. $n / 6 = 9$

B. $6n = 9$

C. $n - 6 = 9$

D. $n + 6 = 9$

Your answer:

Answer Key & Teacher Guide

1. **We Try (notes):** A. $n - 7 = 18$ — *'Seven less than a number' means start with the number and subtract 7: $n - 7 = 18$.*
2. **You Try (notes):** A. $3n = 21$ — *'Three times' means multiply by 3, so the equation is $3n = 21$.*
3. **Try It 1:** A. $3n = 21$ — *'3 times a number' is 3n, equal to 21.*
4. **Try It 2:** A. $8t = 48$ — *Cost per ticket times number of tickets equals total: $8t = 48$.*
5. **Exit Ticket:** A. $n / 6 = 9$ — *'Divided by 6' means division: $n / 6 = 9$. Check: $54 / 6 = 9$ ✓*

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Equation is a math sentence with an equal sign showing both sides are the same.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).